Portfolio

Portfolio

MUHAMMAD
JUMI'AT MOKHTAR

Muhammad Jumi'at Mokhtar

mjmokhtar.cloud
mjmmiat01@gmail.com
(+62)853 4941 6700

EXPERIENCE

2021
Apprenticeship – CV. Saka Persada
2022
Engineer & Instructor – PDWA ULM
2023 - 2025
Engineer – PT. Respati Solusi Teknologi (RESTEK)
2025 - Now
IoT Presales Engineer – PT. Aman Media Interaktif (AMANI Tech)

EDUCATION

2017-2022
Instrumentation Physics
Lambung Mangkurat University
Banjarbaru

INTERESTS

Technical Product & Project Manager (IoT)
System Design & Integration
IoT, Embedded & Firmware Systems
PCB Layout & Circuit Design Engineer
Programming & Scripting
Network Configuration & Protocols

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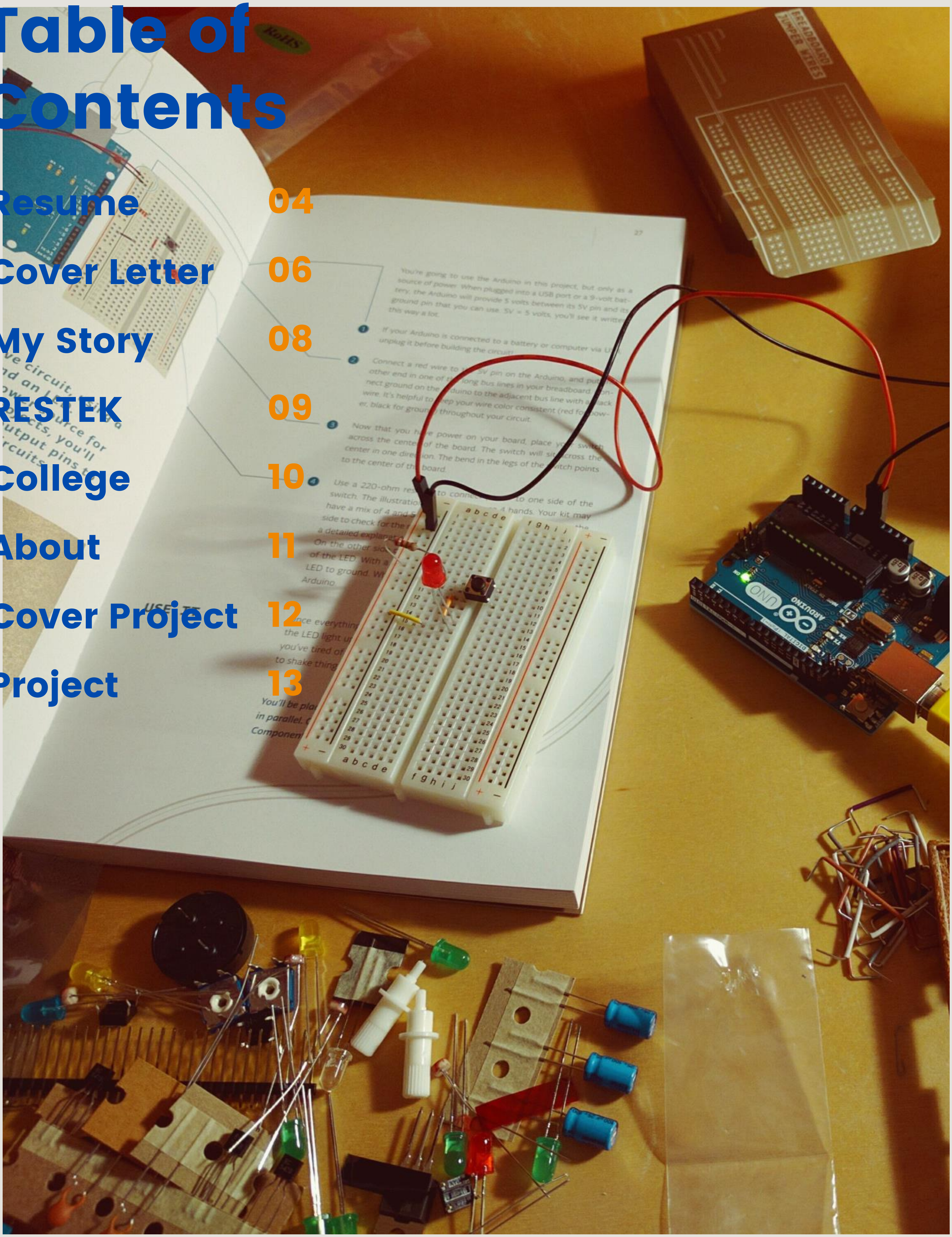


Muhammad Jumi’at Mokhtar
Techincal Product & Project Manager

Email mjmmiat01@gmail.com
Website https://mjmokhtar.cloud

About

I am a Technical Product-Oriented Professional with experience in delivering end-to-end IoT and system integration solutions. I focus on translating real-world operational needs into scalable digital products by defining requirements, aligning stakeholders, and ensuring successful system delivery. With a strong technical background, I bridge engineering execution with product strategy, enabling efficient development, validation, and deployment.

Education

2017 - 2022
Bacherol Science of Pyhsics
Lambung Mangkurat University
- Expertise of Electronics and Instrumentation
- GPA: 3.62/4.00

Training

2023
Project Management
Digital Talent Scholarship
Certified BNSP on: May 30, 2024

Experience

2024 – 2025
System Engineer
PT RESPATI SOLUSI TEKNOLOGI
- Actively engaged with clients to gather feature requirements and manage change requests for the IMAM apps (Integrated Maintenance Asset Management) system, coordinating with developers to align timelines and ensure delivery.
- Participated in business process validation by reviewing and ensuring system implementation matched operational requirements.

2025 – Present
IoT Engineer
PT Aman Media Interaktif
- Led Amani Visual Analytic platform development using OpenCV and YOLO for real-time image and video validation, reducing manual verification time by 60% (processing photos in ~5s and videos in ~6min per file).

Skills

- Product & Project Management: Product Strategy & PRD; Project Execution & Delivery; Agile (Scrum/Kanban); Stakeholder Communication; Technical Product Leadership (IoT Systems)
- Software & Tools: Microsoft Office (Word, Excel, PowerPoint); Collaboration Tools (Jira, Notion, Trello); Development Tools (VS Code, Arduino IDE, STM32CubeIDE, ESP-IDF)
- Programming & Integration : Web Technologies (Node.js, PHP, Python, Golang); API & Data Integration (REST API, Webhooks, MQTT); GUI (C & C++) ; Low-Code & Automation (Node-RED); Version Control Management (Git, Github)
- Systems & Communication: Networking & Protocols (Modbus, TCP, UDP, FTP); Database Systems (MySQL, PostgreSQL)
- Embedded & IoT (Supporting Skills): Microcontrollers (Arduino MCU Family, ESP32, STM32, Raspberry Pi Pico); Mini PC Platforms (Raspberry Pi, LattePanda); Connectivity Technologies (Ethernet, WiFi, BLE, RS485, CAN-BUS, NB-IoT); PLC (Schneider)

Muhammad Jumi'at Mokhtar
Solution Engineer

Email	mjmmiat01@gmail.com
Website	https://mjmokhtar.cloud

About

I am a Solution Engineer with hands-on experience in designing and delivering end-to-end technical solutions that bridge real-world operations with digital systems. My background spans system integration, backend services, API-based communication, and data-driven platforms, with strong foundations in JavaScript, REST APIs, and SQL.

Education

2017 - 2022

Bacherol Science of Pyhsics

Lambung Mangkurat University

- Expertise of Electronics and Instrumentation
- GPA: 3.62/4.00

Sertification

2024

IoT Network Engineer

PT. LSP Telekomunikasi Digital Indonesia

Certified BNSP on: May 30, 2024

Experience

2023 – 2025

Engineer

PT RESPATI SOLUSI TEKNOLOGI

- Integration monitoring and control system of Automatic Train Wash Plant (ATWP) of the Jabodebek LRT based on Modbus TCP/IP
- Making PCB controller hardware and LED Matrix Running Text software
- Designed hardware and create web base monitoring position and parameter of moving trainset

2025 – Present

IoT Engineer

PT Aman Media Interaktif

- The validator ensures the consistency, accuracy, and integrity of critical driving behavior and driver condition data, helping to identify anomalies, validate event triggers, and improve overall system reliability.

Skills

- Software : Microsoft Applications (Word, Excel, PowerPoint); Development Tools (Visual Studio Code, Arduino IDE, STM32CubeIDE, ESP-IDF, Thonny IDE); Operating Systems (Linux, Windows); Collaboration Tools (Jira, Notion)
- Programming & Integration : Backend & Web Technologies (JavaScript, TypeScript, Node.js, PHP, SQL, Python, Golang); API & Data Integration (REST API, Webhooks); GUI (C & C++) ; MicroPython); Low-Code & Automation (Node-RED); Version Control (Git)
- Systems & Communication: Networking & Protocols (MQTT, Modbus TCP/IP, UDP, FTP); Database Systems (MySQL, PostgreSQL)
- Embedded & IoT (Supporting Skills): Microcontrollers (Arduino MCU Family, ESP32, STM32, Raspberry Pi Pico); Mini PC Platforms (Raspberry Pi, LattePanda); Connectivity Technologies (Ethernet, WiFi, BLE, RS485, NB-IoT, CAN-BUS); PLC (Schneider)

Muhammad Jumi'at Mokhtar
IoT, Embedded & Firmware Systems

Email mjmmiat01@gmail.com
Website https://mjmokhtar.cloud

About

Embedded Systems & IoT Developer with a strong physics background and hands-on experience in designing, prototyping, and deploying connected embedded solutions. Skilled in firmware development, sensor integration, and real-time data communication. Proven track record in delivering reliable instrumentation and monitoring systems, with strengths in problem solving and cross-functional collaboration. Seeking to contribute to cutting-edge IoT and embedded system innovation in industrial and research environments.

Education

2017 - 2022
Bacherol Science of Pyhsics
Lambung Mangkurat University
- Expertise of Electronics and Instrumentation
- GPA: 3.62/4.00

Sertification

2024
IoT Network Engineer
PT. LSP Telekomunikasi Digital Indonesia
Certified BNSP on: May 30, 2024

Experience

2023 – 2025
Engineer
PT RESPATI SOLUSI TEKNOLOGI
- Integration monitoring and control system of Automatic Train Wash Plant (ATWP) of the Jabodebek LRT based on Modbus TCP/IP
- Making PCB controller hardware and LED Matrix Running Text software
- Designed hardware and create web base monitoring position and parameter of moving trainset

2025 – Present
IoT Engineer
PT Aman Media Interaktif
- The validator ensures the consistency, accuracy, and integrity of critical driving behavior and driver condition data, helping to identify anomalies, validate event triggers, and improve overall system reliability.

Skills

- Software : Microsoft Applications (Word, Excel, Powerpoint); IDE Tools (Arduino IDE, STM32CubeIDE, ESP-IDF, Thonny IDE, Visual Studio Code); Autodesk Applications (Eagle PCB); Operation System (Linux and Windows) Collaboration Tools (Jira, Notion)
- Programming IoT Developer : Web programming for Internet of Things (JavaScript, TypeScript, Bun, Vue.js, Node.js, Node-RED, php, SQL, Python, Golang); Microcontroller (C & C++); Protocols Peripherals: (UART, SPI, I2C) N/W: (MQTT, Restful API, Modbus RTU & TCP/IP, UDP, FTP); Version Control (Git); Database Systems (MySQL, PostgreSQL)
- Embedded Systems Engineer : Mini PC (Lattepanda, Raspi); Microcontroller (Arduino/AVR MCU Family, STM32 MCU, Espressif MCU, Raspi pico); addition device (nRF24L01, SIM Card Modul, LoRa, RS485, RS232, CAN-BUS, BLE, WiFi, NB-IoT, Ethernet); PLC Schneider

Muhammad Jumi’at Mokhtar
Technical Product & Project Manager

Dear HR Team,

My name is Muhammad Jumi’at Mokhtar, and I hold a Bachelor’s degree in Physics with a specialization in Electronics and Instrumentation from Universitas Lambung Mangkurat. I am writing to express my interest in the Solution Engineer position, with a strong focus on evolving into product-oriented roles.

I have experience in developing end-to-end IoT and system integration solutions by translating real-world operational needs into structured product requirements and scalable systems. I actively contribute not only to technical implementation, but also to defining features, validating user needs, and ensuring that solutions deliver consistent value in real-world environments.

At PT. Respati Solusi Teknologi (RESTEK), I was involved in building monitoring and control systems for transportation infrastructure, while also engaging with clients to gather requirements, manage feature changes, and validate system alignment with business processes. This experience strengthened my ability to connect technical execution with product needs.

Currently, at PT. Aman Media Interaktif, I contribute to IoT and analytics solutions by collaborating with engineering and analytics teams to design validation logic, ensure data accuracy, and support solution discussions. I am also involved in early-stage discussions to help shape solution approaches based on user and business requirements.

With a strong technical background and a growing product mindset, I am motivated to bridge engineering, user needs, and business goals—ensuring that every system built is not only functional, but also valuable, scalable, and aligned with product objectives.

Thank you for your time and consideration. I would welcome the opportunity to further discuss how my experience aligns with your needs.

Sincerely,
Muhammad Jumi’at Mokhtar

Muhammad Jumi’at Mokhtar
Solution Engineer

Dear HR Team,

My name is Muhammad Jumi’at Mokhtar, and I hold a Bachelor’s degree in Physics with a specialization in Electronics and Instrumentation from Universitas Lambung Mangkurat. I am writing to express my interest in the Solution Engineer position, where I can contribute my experience in system integration, backend development, and customer-facing technical solutions.

I have hands-on experience designing and delivering end-to-end systems that integrate devices, backend services, databases, and APIs. In my previous roles, I worked closely with cross-functional teams to translate operational and business requirements into reliable technical solutions, covering system design, implementation, testing, and deployment.During my time at PT. Respati Solusi Teknologi (RESTEK), I was involved in developing monitoring and control systems for large-scale transportation infrastructure, supporting real-time data acquisition and system reliability in production environments. Currently, at PT. Aman Media Interaktif, I support IoT and analytics solutions by coordinating with analytics and engineering teams to design validation logic and ensure data accuracy, consistency, and system reliability—while also contributing to presales and solution discussions.

In addition to my technical background, I am comfortable communicating complex technical concepts to non-technical stakeholders. My experience in project coordination, documentation, and collaboration tools has strengthened my ability to work effectively with customers and internal teams to deliver impactful solutions.

I am highly motivated, adaptable, and eager to continue growing as a Solution Engineer in a fast-paced SaaS and technology-driven environment. I believe my background in system integration and real-world deployment will allow me to contribute meaningfully to your team.

Thank you for your time and consideration. I would welcome the opportunity to further discuss how my experience aligns with your needs.

Sincerely,
Muhammad Jumi’at Mokhtar

Muhammad Jumi’at Mokhtar
IoT, Embedded & Firmware Systems

Dear HR Team,

My name is Muhammad Jumi’at Mokhtar, and I hold a Bachelor’s degree in Physics with a specialization in Electronics and Instrumentation from Universitas Lambung Mangkurat. I am writing to express my interest in an Embedded Systems / Firmware Engineer position, where I can contribute my experience in embedded development, system integration, and real-world deployment.

I have hands-on experience in designing, developing, and deploying embedded systems for monitoring, control, and data acquisition applications. My background includes firmware development on microcontroller platforms, sensor integration, system calibration, and communication protocols, with practical exposure to both standalone and connected embedded systems.

During my professional experience at PT. Respati Solusi Teknologi (RESTEK), I worked on embedded and backend-integrated systems for transportation infrastructure, supporting real-time monitoring and control in production environments. I was involved in developing controller hardware, implementing firmware logic, and ensuring system reliability across deployed units. Currently, at PT. Aman Media Interaktif, I work on IoT-based systems and data validation workflows, collaborating with analytics and engineering teams to improve system accuracy, reliability, and operational insight.

In addition to my technical skills, I am accustomed to working in cross-functional teams and adapting to project requirements in dynamic environments. My academic background and involvement in research and teaching activities have also strengthened my analytical thinking, documentation, and communication skills.

I am motivated to continue growing as an Embedded Systems Engineer and to contribute to projects that require reliable, well-engineered solutions. I believe my combination of academic foundation and industry experience would allow me to add value to your engineering team.

Thank you for your time and consideration. I look forward to the opportunity to further discuss how my experience aligns with your needs

Sincerely,
Muhammad Jumi’at Mokhtar

Technical & Engineer Story

2020

Grant Research Sawit BPDPKS

Badan Pengelola Dana Perkebunan Sawit

Creating and calibrating temperature gauges to measure palm heating temperature of palm oil in a dump truck

2021

Apprenticeship

CV. Saka Persada

Working on the manufacture of telemetry tools and systems at the Tapin dam, Kalimantan Selatan, Indonesia

2022

Engineer & Instructor

PDWA ULM

Participated in a lecturer-led research program as an engineer in agricultural monitoring projects, while also delivering Arduino and robotics training for high school students.

2017 - 2022

Bachelor of Science

Lambung Mangkurat University

● Expertise of Electronics and Instrumentation

● Member of the Physics Student Association, Islamic Spirituality, and Student Representative Council

● 1st Place in Physics Competition/Olympic at University Level in 2021 and Bronze at National Level in 2020

● Practicum Assistant: Basic Physics, Sensor Systems, Measurement Systems, Digital Electronics, Industrial Instrumentation, Microprocessor and Interfacing

● GPA: 3.62/4.00

IoT, Embedded & Firmware Systems, Solution Engineer

2025 - Present

IoT Presales Engineer

PT Aman Media Interaktif (AMI)

Core Responsibilities

- Proficiency in programming languages like C, C++, Python, Node.js, Golang for backend development.
- Collaborated with clients to understand IoT requirements and design tailored technical solutions
- Prepared technical proposals, solution architectures, and product demonstrations
- Conducted Proof of Concept (PoC) to validate feasibility and performance of IoT solutions
- Supported IoT device integration, ensuring reliable data communication and system interoperability
- Worked closely with sales, solution architects, and engineering teams to align solutions with business needs
- Produced technical documentation covering system architecture, integration flows, and use cases
- Applied best practices in IoT security, data management, and scalable deployment
- Provided ongoing technical support and guidance throughout project lifecycles

2024

IoT Network Engineer

PT. LSP Telekomunikasi Digital Indonesia

- Training provided by Floatway Systems, covering integration, testing, operation, OTA firmware installation, and security for IoT devices. Also Embedded Systems, and IoT Architecture.
- Certified BNSP on: May 30, 2024

2023 - 2025

Engineer

PT Respati Solusi Teknologi (RESTEK)

- Designed and delivered end-to-end monitoring and control systems for LRT infrastructure
- Integrated backend services, embedded devices, and web-based monitoring platforms
- Supported real-time data acquisition, validation, and reliable deployment in production environments
- Led technical discussions and application development coordination, including requirement clarification, solution alignment, and cross-team collaboration to ensure timely and reliable system delivery.

Manager Story

Techincal Product & Project Manager

2026

Product & Project Management

Self Project

- From a hands-on engineer to a hybrid builder-PM — managing real projects end-to-end, from embedded hardware to stakeholder documentation, all at once. From executing other people's visions to building my own: J2ME Retro Archive, ePaper conference badge, and Toaster Co.

2026

Project Management

PT Aman Media Interaktif

- I started as an engineer who builds things — then realized the most important thing to build is the right thing. Now I do both.

2026

Product & Project Management

PT Jay Jay Edukasi

- Led cross-functional delivery of IoT and embedded systems projects, bridging technical execution with stakeholder alignment and documentation.
- Managed full project lifecycle using PMBOK-aligned practices — Gantt, Risk Register, RACI, S-Curve — while remaining hands-on as both engineer and developer.

Muhammad
Jumi'at Mokhtar
Student Hustler

Phone number:	(+62) 853-4941-6700
Email	mjmmiat01@gmail.com
Website	linktr.ee/mjmokhtar

Organization at Collage

- Member of the Physics Student Association (HIMAFI FMIPA ULM) in 2018
- Member of the Islamic Spirituality (FSI Ulul Albab FMIPA ULM) in 2018
- Division Coordinator of Islamic Spirituality (FSI Ulul Albab FMIPA ULM) in 2019
- Head of the General Election Commissions (KPM FMIPA ULM) in 2018 – 2019
- Chairman of the Legislative Commissions Student Representative Council (DPM FMIPA ULM) in 2020
- Head of the instrumentation community in 2020

Practicum Assistant

- Basic Physics in 2018
- Sensor Systems in 2020 & 2021
- Digital Electronics in 2020
- Measurement Systems in 2021
- Industrial Instrumentation in 2021
- Microprocessor and Interfacing in 2021

Achievements

- ONMIPA University Runner up in 2019
- KNMIPA University Runner up in 2020
- ONMIPA BKS PTN Barat Bronze in 2020
- KNMIPA National participant 2020
- KNMIPA University Champion in 2021

Lecturer Project

- Grant Research on Oil Palm: Conducted under the BPDPKS research grant. Developed and calibrated temperature measurement tools for monitoring oil palm heating inside dump trucks (2020).
- Built IoT-based environmental monitoring systems using ESP8266/ESP32 with 5+ sensors (light, soil temp/moisture, air temp, humidity).Developed web dashboard (PHP, MySQL) with REST API & MQTT, achieving <2s latency and ~99% device uptime.
- Delivered Arduino-based robotics workshops to 30+ students and authored a beginner-friendly Arduino learning module for MAN 4 Banjar School.
- **Undergraduate thesis**: Designed and built an IoT-based agricultural monitoring system with multi-sensor data acquisition and real-time PC dashboard, including automated logging for analysis and validation.

Muhammad Jumi’at Mokhtar As Engineer at RESTEK

Phone number:	(+62) 853-4941-6700
Email	mjmmiat01@gmail.com
Website	https://cv-ats.mjokhtar.cloud/

Project in RESTEK

- **EAMMS ATWP**: Developed backend software for monitoring and controlling the Automatic Train Wash Plant (ATWP) of LRT Jabodebek, integrating PLC systems via Modbus TCP/IP protocol. reducing manual oversight by 40%. Built server-side applications using JavaScript and PostgreSQL for real-time data acquisition 24/7 operation uptime, and process control.
- **Running Text**: Designed and developed PCB controller modules and firmware for RGB LED Matrix Running Text displays, programmed using Arduino IDE, deployed in >50 train units in LRT Jabodebek, Commuter Line, and Feeder trains KCIC. Improving system stability by 30%.
- **GPS Tracker**: Designed an ESP32–Raspberry Pi–based monitoring system integrating GPS, accelerometer, and magnetometer sensors, using UDP and 4G connectivity for real-time trainset tracking enabling continuous remote with <2 s update intervals and 98% data reliability.

Beyond the 9 to 5

- **Logger computing**: Developed a Python-based parser converting .BIN files from the Train Information System (TIS) into CSV datasets containing physical parameters, GPS coordinates, and train conditions, enabling real-time cloud monitoring and reducing manual data analysis time by 75%.
- **CCD Train**: Integrated Schneider M221 PLC with ultrasonic sensors and CCTV to enable remote monitoring of the Current Collector Device (CCD) Train, reducing on-site inspection frequency by 60% and improving maintenance efficiency.
- **Middleware of measurement system**: Developed a real-time weight-sensor data processing system with FTP image backup and camera integration, enhancing data traceability and reducing manual reporting time by 40%.

Muhammad Jumi’at Mokhtar As IoT Presales Engineer at AMANI Tech

Phone number:	(+62) 853-4941-6700
Email	mjmmiat01@gmail.com
Website	https://cv-ats.mjokhtar.cloud/

Project in AMANI Tech

● **ADAS and DSM Validator System**: The validator ensures the consistency, accuracy, and integrity of critical driving behavior and driver condition data, helping to identify anomalies, validate event triggers, and improve overall system reliability.
● Led Amani Visual Analytic platform development using OpenCV and YOLO for real-time image and video validation, reducing manual verification time by 60% (processing photos in ~5s and videos in ~6min per file).
● Collaborated with AI and infrastructure teams to optimize data pipeline throughput, improving system reliability and event validation accuracy by 25%.
● **Desktop Middleware BatteryCharger App**: Developed a dekstop middleware system for Battery charging. using serial comunication protocol to convert byte data from old apps to new device battery charging using protocol SCPI.

Beyond the 9 to 5

● **Freelance**: Provide one-on-one private tutoring in embedded systems, covering microcontroller programming, circuit design, IoT protocols, and real-world hardware integration for students and professionals.
● **J2ME Retro Archive Platform**: Built a web-based archival platform for classic J2ME mobile games, featuring structured content management, fast client-side rendering, and scalable backend services for handling large digital collections. (https://j2me-retro.mjmokhtar.cloud) (Vue.js, Bun, TypeScript, PostgreSQL).

Muhammad Jumi’at Mokhtar Who I Am

Phone number:	(+62) 853-4941-6700
Email	mjmmiat01@gmail.com
Website	linktr.ee/mjmokhtar

About

One of my strengths is my disciplined nature and strong adherence to ethical values in all situations. I am diligent, proactive in learning, and strive to complete tasks quickly and accurately. Additionally, I am optimistic, honest, and responsible, with a solid ability to work effectively in teams. My communication skills are strong, allowing me to collaborate well with others and solve problems efficiently.

However, I recognize that I have room for improvement in public speaking, as I tend to lack confidence when addressing large groups. Additionally, I sometimes find it challenging to take on leadership roles.

Training Certificate

- Project Management by Coursera and KOMINFO on august 2023
- Arduino : Electronics circuit, PCB Design & IOT Programming on 10/08/2023 taught by Piyush Charpe at Udemy
- Bootcamp Advance IoT for IoT Engineering Batch 1 by Edspert.id on september and november 2023
- Python Programming by Cisco and Digitalen KOMINFO on January 2024
- Golang ProClass (January 2024) by Harisenin.com
- IoT Network Engineer Completed training provided by Floatway Systems prior to certification by PT. LSP Telekomunikasi Digital Indonesia

Language

English	C1/C2
Indonesia	Native

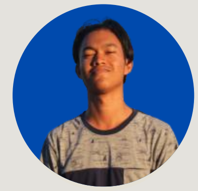
Hobbies

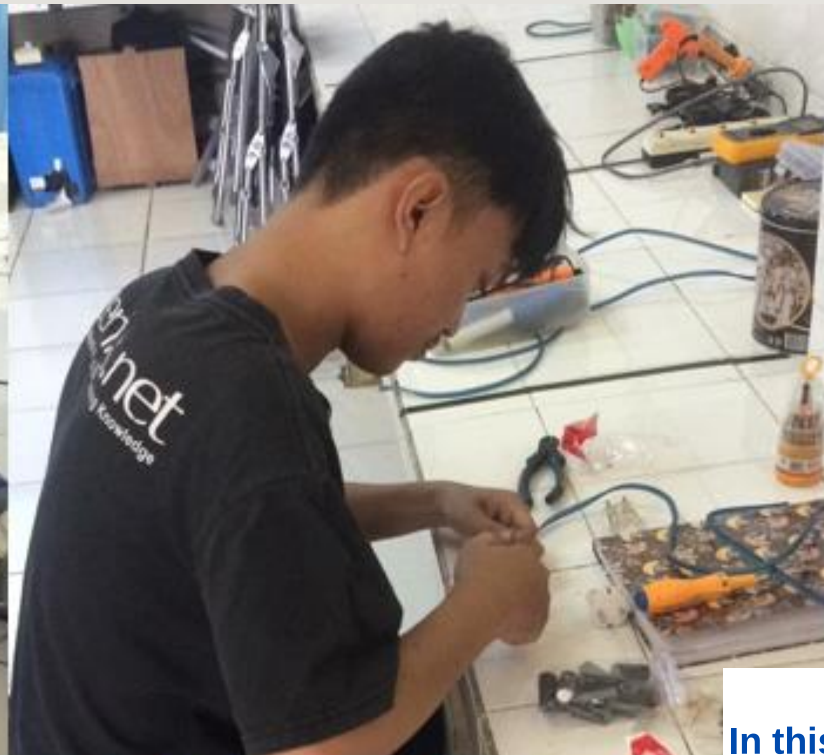
Programming

Hacking

Waching Movie

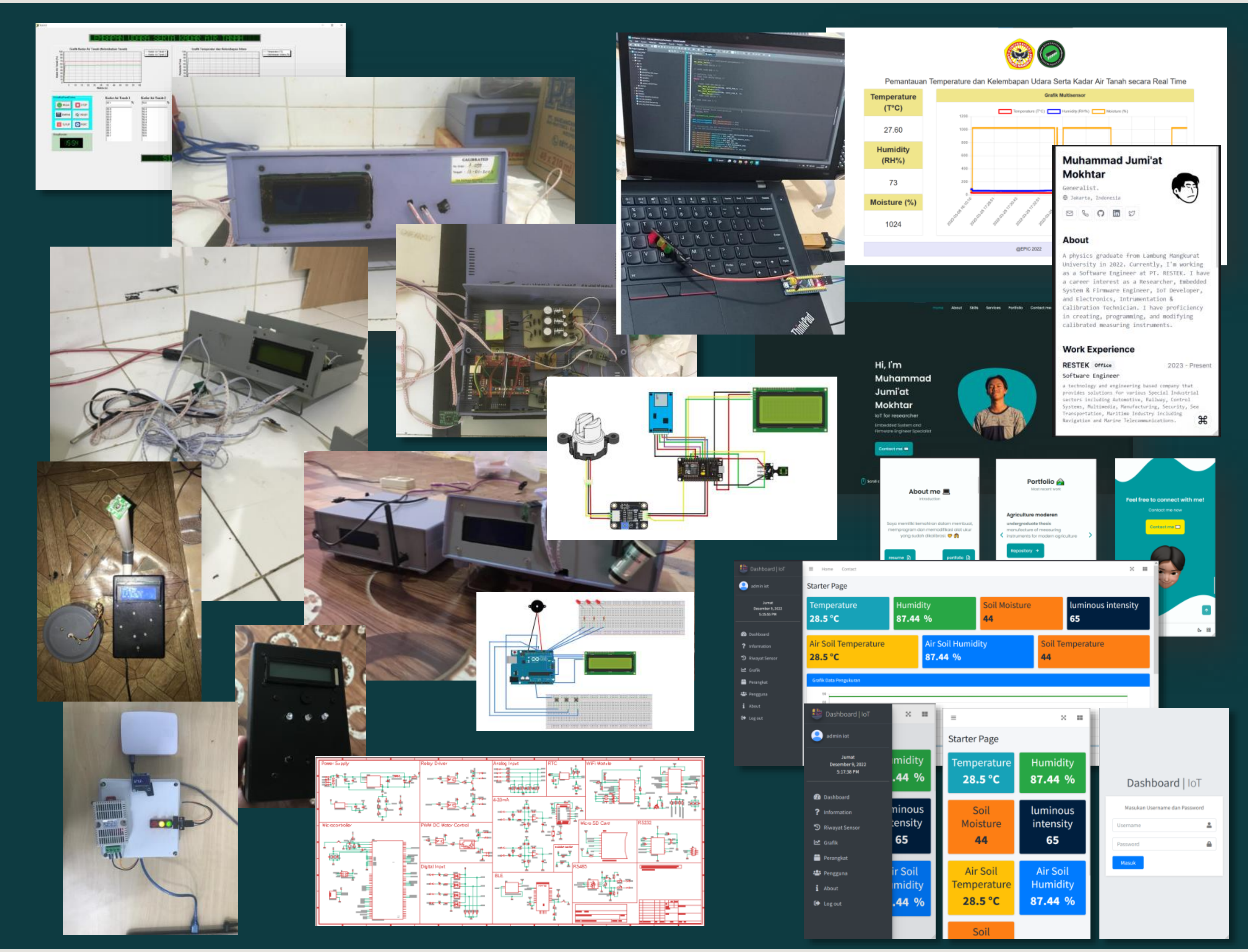
IoT, Embedded & Firmware Systems, Solution Engineer





In this portfolio I will explain the achievements I have achieved during college and also explain the project of designs and products that I have made, both in the form of hardware and software

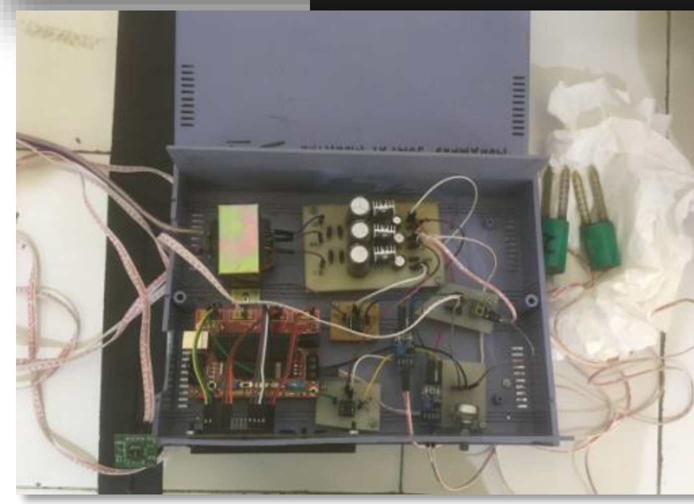
IoT, Embedded & Firmware Systems, Solution Engineer



Project

I was involved in a lot of research on making hardware and software measuring tools, namely making measuring tools Arduino-based, raspberry pi pico-based, ATMega 16A-PU modules and Espressif series NodeMCU ESP8266 & ESP32 and making applications based on Borlan Delphi 7 and Web-based. The application and programming language that I use is Arduino IDE and ESP-IDF (C and C++) and Thony IDE (Micropython)

Manufacture Of A Measuring Instrument For Soil Moisture, Luminous Intensity, For Tomato Plants On Paranet As Well As Temperature And Humidity Environment Based Microcontroller ATMega16A-PU.

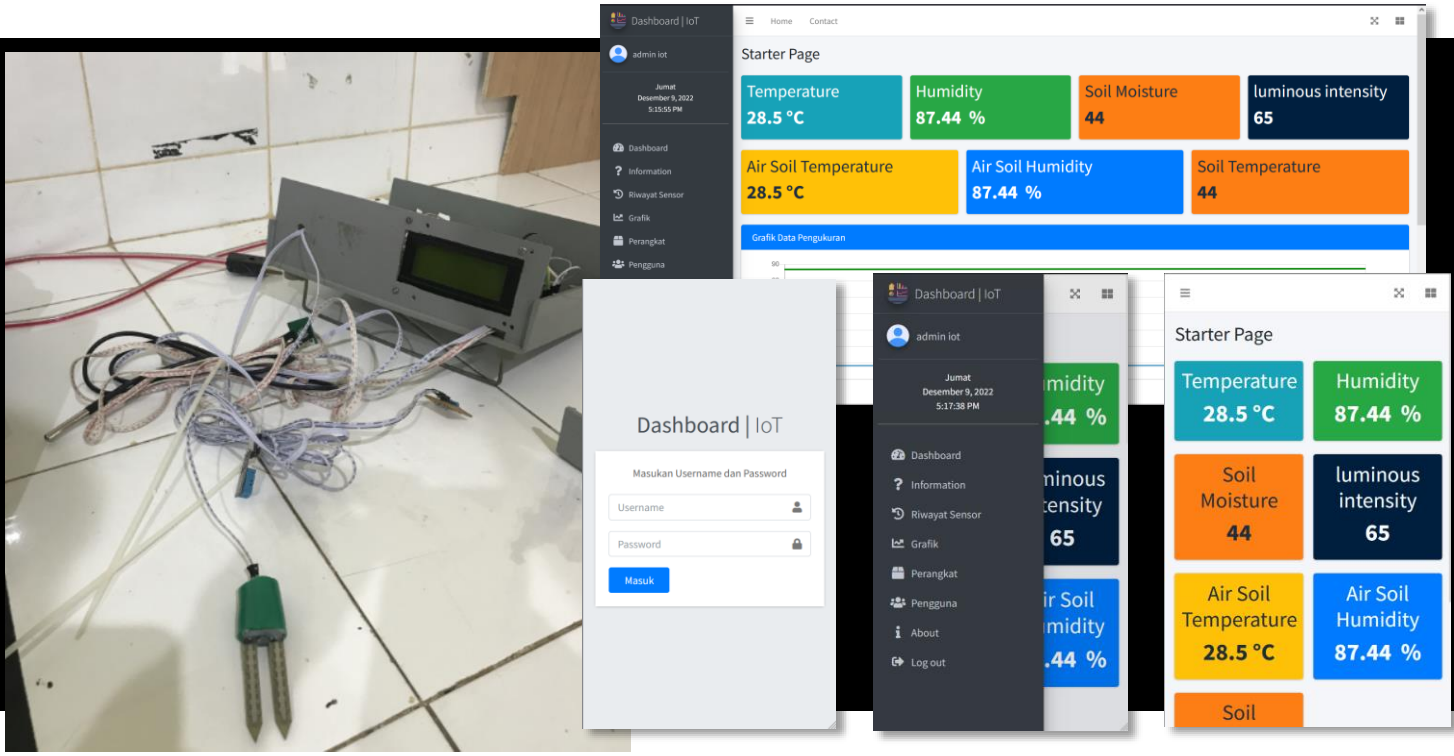


What i'am doing in this project ?

- Worked on date: 3rd March – 14th July 2022
- Making instruments for measuring light intensity, soil moisture content, temperature and humidity using the ATMega 16A-PU microcontroller using the BASCOM IDE application to program the microcontroller and GUI interface using the Delphi 7.0 application
- Characterize sensors for light intensity, soil moisture content and temperature and humidity and calibrate the measuring instrument as a whole
- Create a PC-based monitoring system and record in Excel in real time
- Presented this project at the national seminar on physics and its applications 2022 (SENFIT) at 26th November 2022

Real Time Monitoring of Environmental Parameters of Wetland Agriculture Using Web-Based NodeMCU ESP32

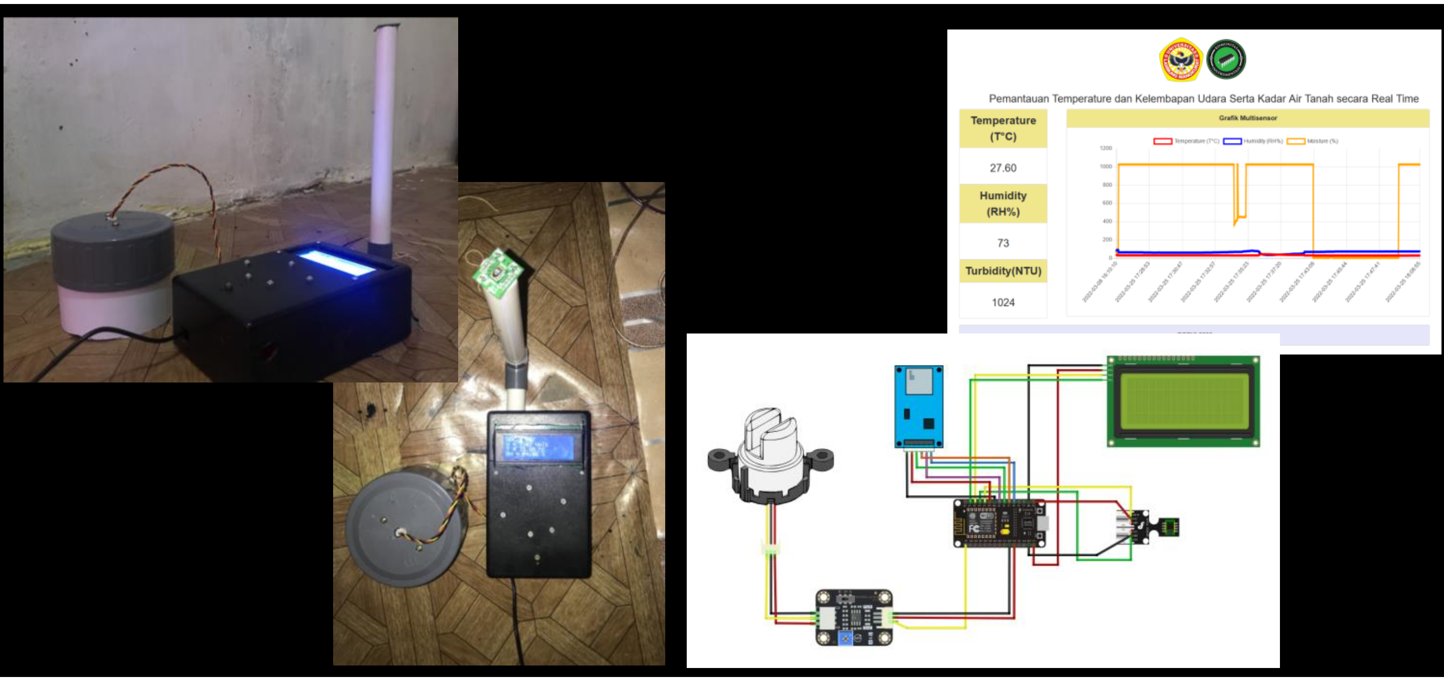
IoT, Embedded & Firmware Systems, Solution Engineer



What i'am doing in this project ?

- Worked on date: 3rd February 2022 – 31th January 2023
- Making instruments for measuring light intensity, soil temperature, soil water content, air temperature and humidity, and underground air temperature and humidity using the NodeMCU ESP32 microcontroller using the Arduino IDE application to program the microcontroller
- Characterize sensors for light intensity, soil temperature, soil water content, air temperature and humidity, and underground air temperature and humidity and calibrate the measuring instrument as a whole
- Create an IoT-based monitoring system and record in MySQL in real time
- Made Responsive design Admin web-based by using HTML, CSS, Javascript Ajax, php native framework

Monitoring Water Turbidity As Well As Temperature And Humidity Environment Using Web-Based NodeMCU ESP8266

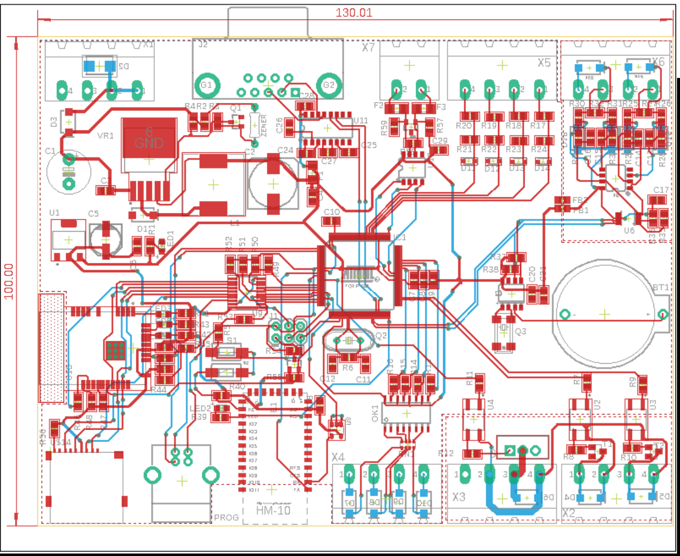


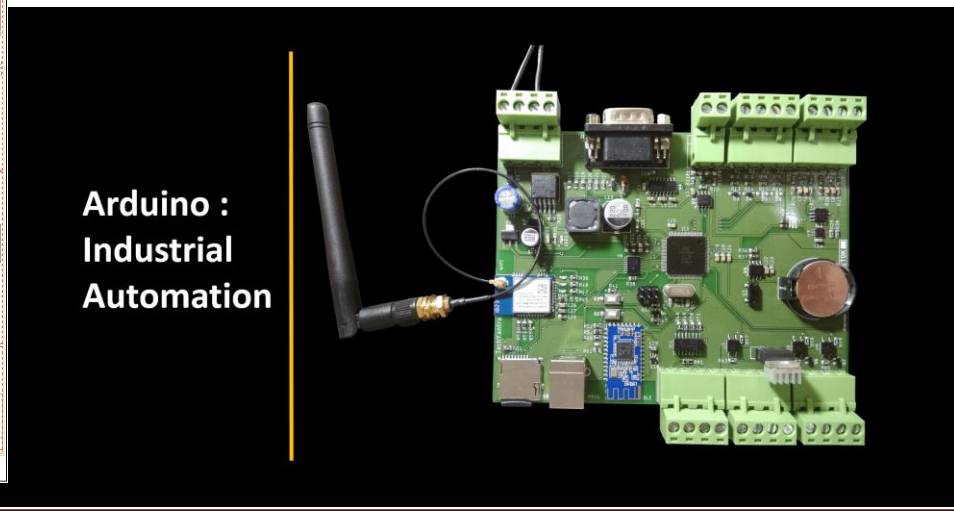
What i'am doing in this project ?

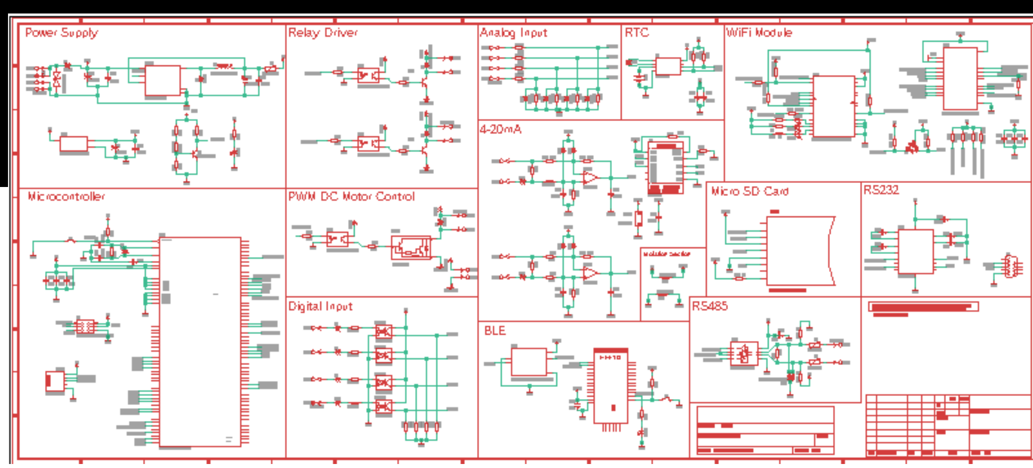
- Worked on date: 1st February – 28th March 2023
- Making instruments for analog sensors turbidity, air temperature and humidity, using the ESP8266 using the Thony IDE (micropython) application to program the microcontroller
- Characterize sensors for water turbidity, air temperature and humidity Enviroment, calibrate the measuring instrument as a whole
- Create an IoT-based monitoring system and record real time
- Data storage is carried out online using the MySQL phpmyadmin database and offline using the SD Card module with *.txt format
- Create Responsive design simple monitoring web-based by using HTML, CSS Bootstrap, Javascript Ajax, php basic

IoT, Embedded & Firmware Systems, Solution Engineer

Making schematic and PCB designs



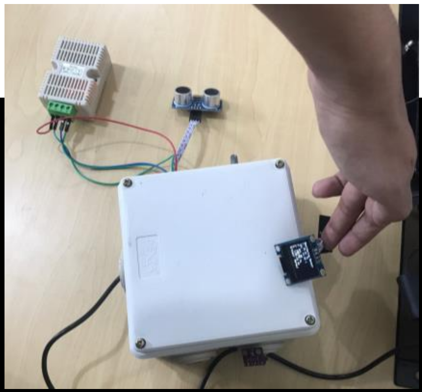


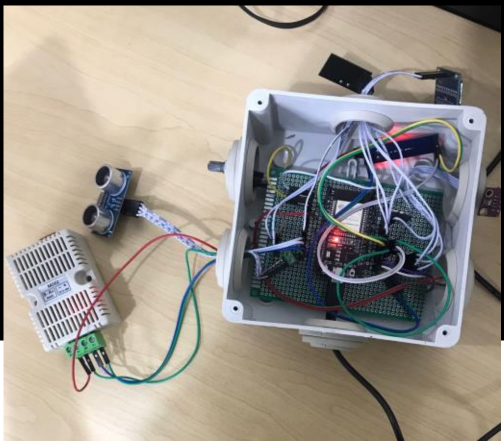


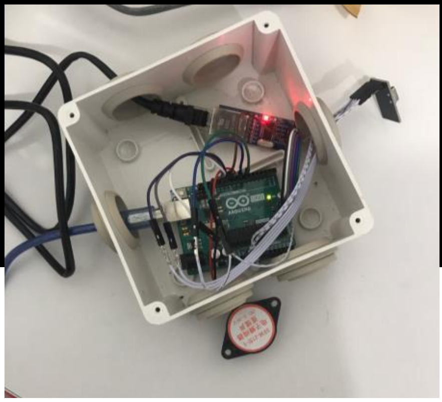
What i'am doing in this project ?

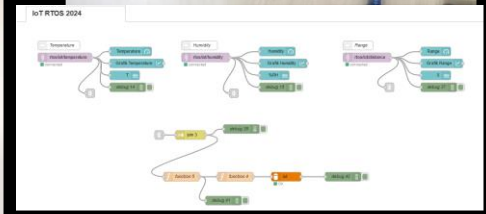
- Worked on date: 1st Juni – 10th August 2023
- Making schematic and pcb desain instruments for analog and digital sensor, digital controler, motor dc control, relay control, serial comunitaction, rs485, rs232, ble, rtc, wifi, sd card, using EAGLE PCB
- Using microcontroller AVR ATMega2560
- Learn how to program it using arduino ide

Making Early warning Systems







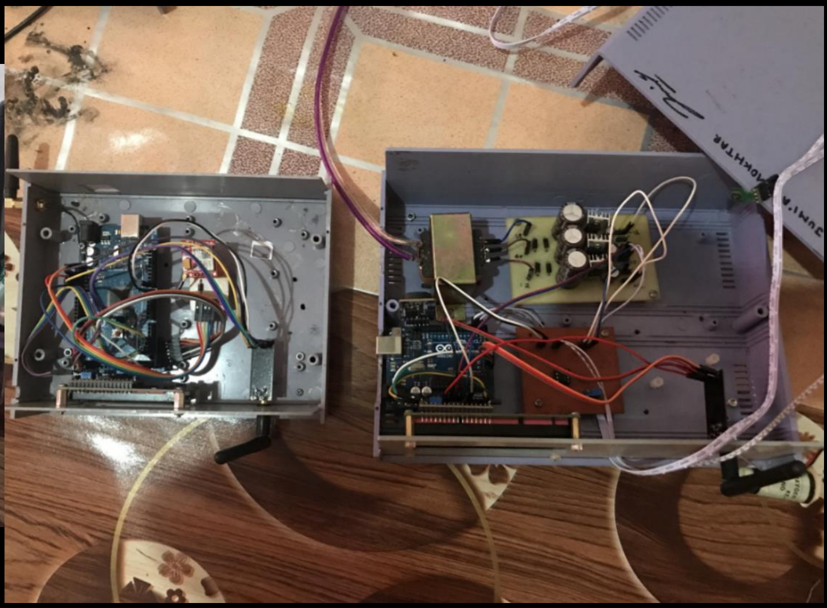


What i’am doing in this project ?

- Worked on date: 1st November – 20th December 2023
- The IoT Early Warning System uses an ESP32 as the transmitter and Arduino Uno as the receiver. It detects and transmits early warning signals for various environmental parameters or events.
- Communication between ESP32 transmitter and Arduino Uno receiver occurs via wireless protocols like nRF24L01+.
- The Early Warning System can be deployed for home security, industrial monitoring, agriculture, or natural disaster detection, adapting to various sensor configurations and warning criteria.
- The project involves utilizing an ESP32 microcontroller running FreeRTOS to directly transmit data to a server.

Monitoring System of Oxygen Levels, Temperature and Humidity in weatland using IoT-based Wireless Communication the MQTT Protocol

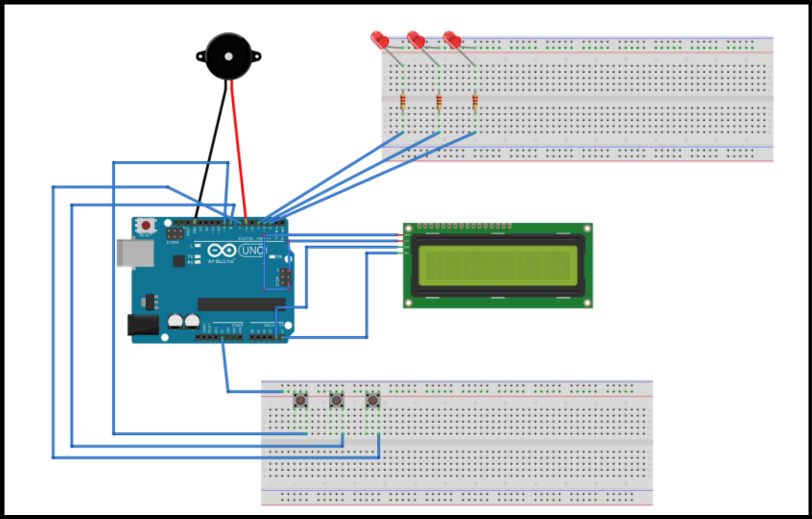
IoT, Embedded & Firmware Systems, Solution Engineer



What i'am doing in this project ?

- Worked on date: 26th November 2022 – 31th January 2023
- Make an instrument for measuring oxygen levels, as well as temperature and humidity using the Arduino Uno microcontroller and SIM 800L for Conected to internet, also using the Arduino IDE application to program the microcontroller and GUI interface using Android
- It consists of 2 devices: the first device is for a system for measuring oxygen levels, as well as temperature and humidity, the second device is for data loggers, data transfers and monitoring systems in real time based on IoT.
- Data storage is carried out online using the MySQL database and offline using the SD Card module with *.txt format
- 2 devices communicate data transfer via nRF24L01+ then data is sent using the MQTT protocol and MySQL

Mood Box Arduino-Based



What i'am doing in this project ?

- Worked on date: 15th December 2022 – 9th January 2023
- Make an Arduino-based mood box with components including a display unit (LCD) to display text, a buzzer to produce the desired sound, a push button and an LED for additional accessories
- The LCD displays words such as congratulation cards and etc
- The buzzer produces the desired song tone
- Push button to adjust LCD display, desired buzzer
- LED as an additional accessory

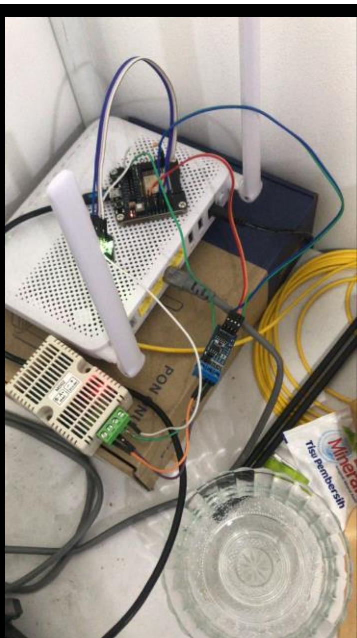
STM32 W550 ENC28J60 LWIP Ethernet System

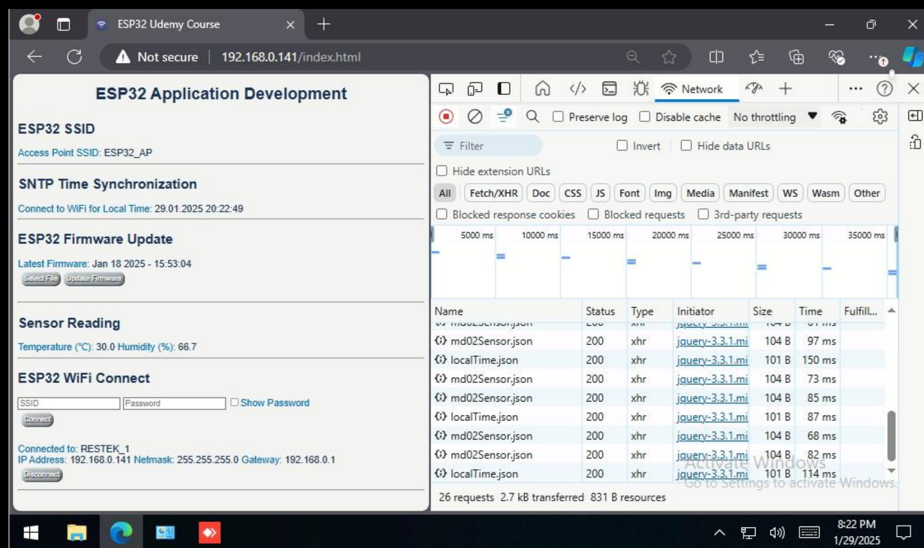
IoT, Embedded & Firmware Systems, Solution Engineer

What i'am doing in this project ?

- Worked on date: 14th – 30th November 2024 (ENC28J60); 1st – 15th Desember 2025 (W5500)
- Integrated STM32F401 microcontroller with the ENC28J60/W5500 Ethernet module for network communication.
- Implemented the LWIP protocol stack to enable TCP/IP communication over Ethernet ENC28J60.
- Developed firmware to handle UDP and TCP protocols for data transmission and reception.
- Optimized resource usage on the microcontroller to ensure efficient system performance.
- Conducted hardware and software integration, including testing and debugging Ethernet communication.
- Create a two-way private UDP-based chat application from a laptop to a device (W5500). The laptop uses a packet sender application, and the device can send back messages using a web server interface.

ESP32 web server dashboard monitoring





What i’am doing in this project ?

- Worked on date: 24th January – 25th March 2025
- Develop WiFi-based applications using ESP-IDF, including an HTTP server and a web page with HTML, JavaScript, and CSS.
- Integrate components and sensors using libraries that support GPIO, PWM, RGB LED control, and button interrupts.
- Implement modular programming and task management with FreeRTOS using message queues, event groups, and semaphores.
- Develop and debug ESP32 applications using ESP-IDF in Eclipse, including building, flashing, and monitoring.
- Utilize SNTP for time synchronization, store data in Non-Volatile Storage (NVS), and enable OTA firmware updates.

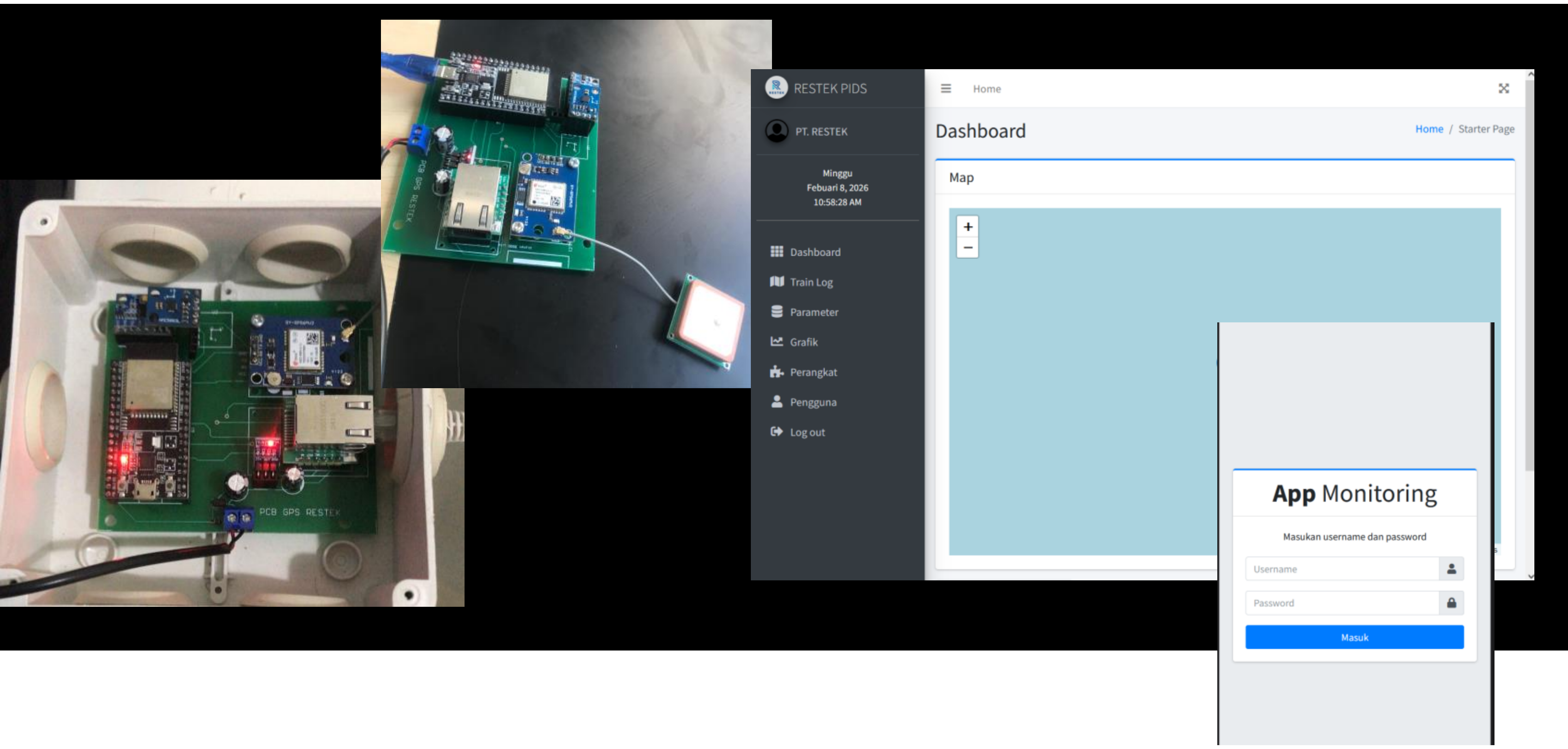
Running Text FDI Train

IoT, Embedded & Firmware Systems, Solution Engineer

What i’am doing in this project ?

- Worked on date: 14th May – 30th November 2024
- Designed hardware PCB and firmware for a running text display unit using dot matrix LED panels on trains as front display information system
- Implemented the LWIP protocol stack to enable TCP/IP communication over Ethernet.
- Developed firmware to handle UDP protocols for data transmission and reception.

GPS Tracker
Train Monitoring

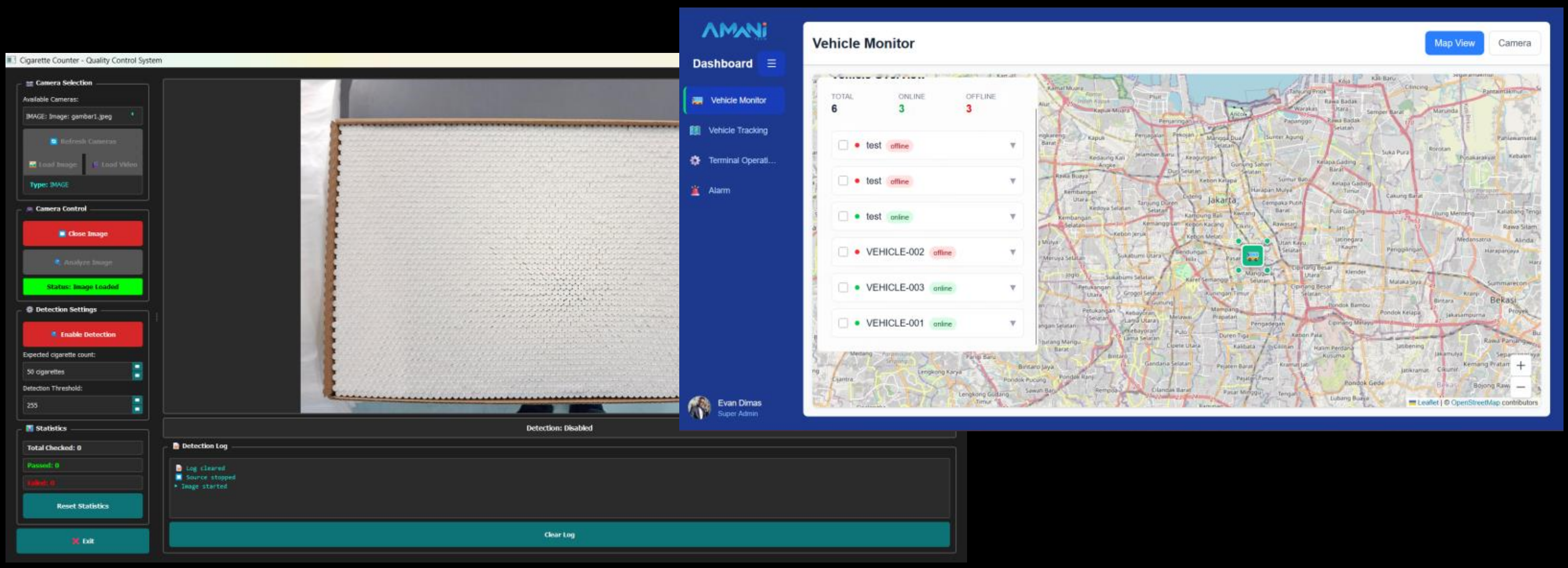


What i'am doing in this project ?

- Worked on date: 1st January – 6th April 2025
- Designed an ESP32–Raspberry Pi–based monitoring architecture for real-time trainset tracking
- Integrated GPS, accelerometer, and magnetometer sensors for continuous position and motion data acquisition
- Implemented UDP-based data transmission over 4G networks to enable low-latency remote monitoring
- Achieved reliable real-time updates with <2-second refresh intervals and 98% data transmission reliability

Visual Analytic Dashboard

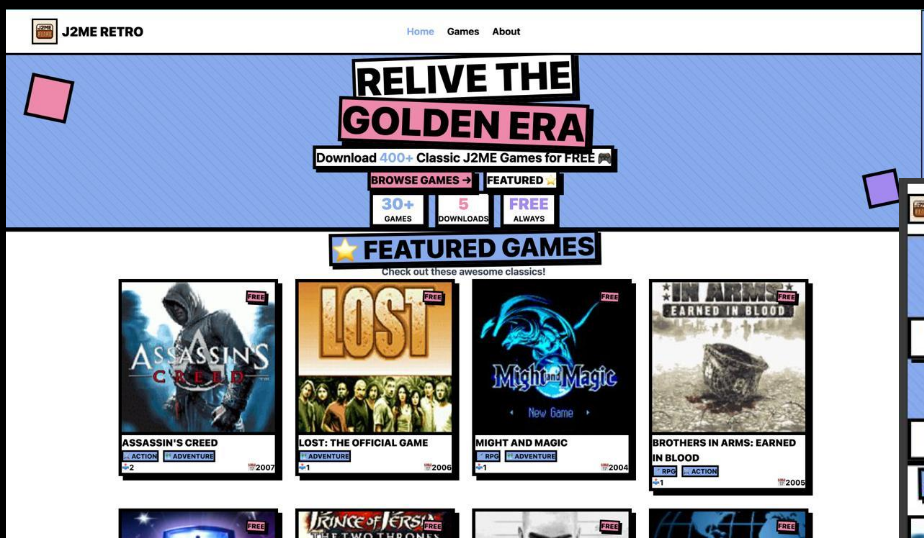
IoT, Embedded & Firmware Systems, Solution Engineer

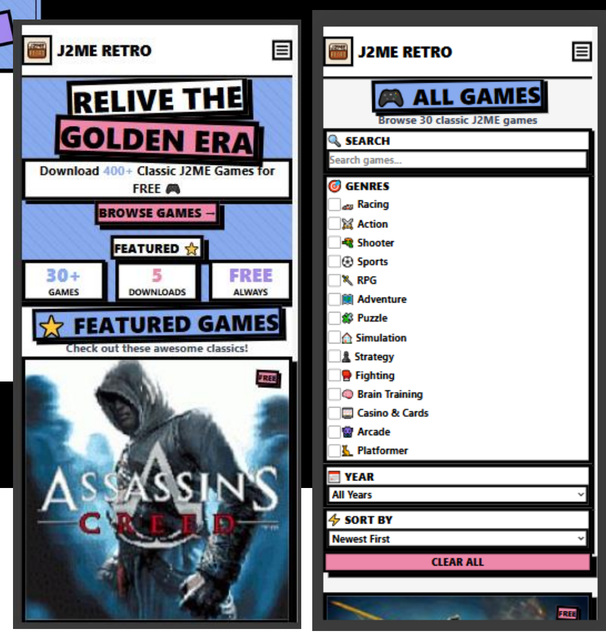


What i'am doing in this project ?

- Worked on date: 14th May – 30th July 2025
- The validator ensures the consistency, accuracy, and integrity of critical driving behavior and driver condition data, helping to identify anomalies, validate event triggers, and improve overall system reliability.
- Led Amani Visual Analytic platform development using OpenCV and YOLO for real-time image and video validation, reducing manual verification time by 60% (processing photos in ~5s and videos in ~6min per file).
- Collaborated with AI and infrastructure teams to optimize data pipeline throughput, improving system reliability and event validation accuracy by 25%.

J2ME RETRO



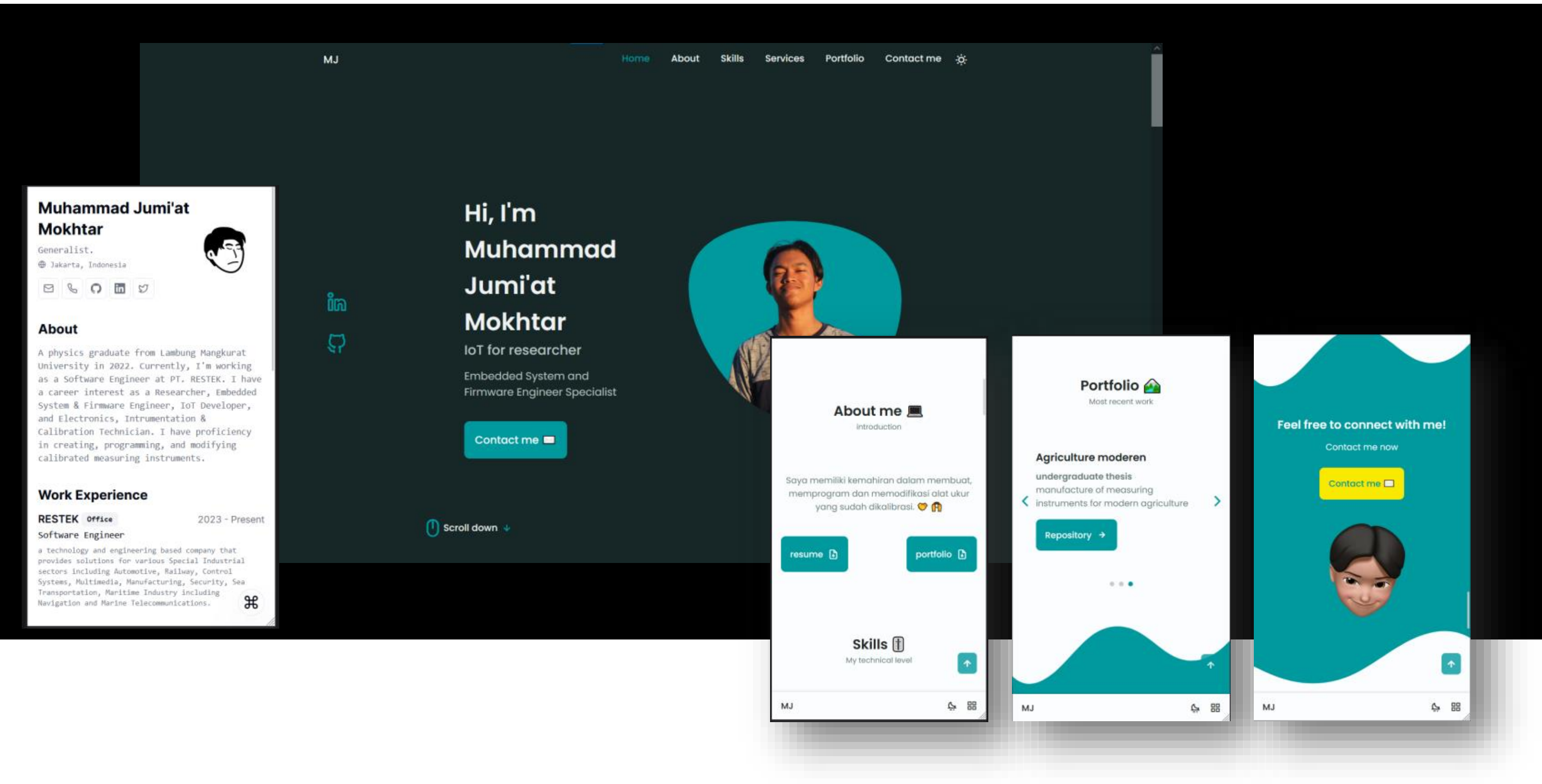


What i'am doing in this project ?

- Worked on date: 1st January – 25th March 2026
- Built full-stack web platform using Vue 3, Hono, Bun runtime, and PostgreSQL with TypeScript for type safety and modern development workflow.
- Implemented JWT authentication system for admin dashboard with secure password hashing, token-based sessions, and role-based access control.
- Developed RESTful API with Drizzle ORM including CRUD operations, file upload handling (JAR/images), and signed download tokens with expiry.
- Created admin dashboard with game management (create, edit, delete), multi-file upload, search/filter, and real-time statistics display.
- Designed public website with neobrutalism UI, advanced filtering (genre, year, search), screenshot galleries, and secure download system.
- Deployed with PM2 process manager and Cloudflare Tunnel for production hosting with auto-restart, log rotation, and zero-downtime updates.
- Achieved 400+ games capacity, automated database backups, optimized PostgreSQL queries with indexes, and CORS-secured API endpoints.

IoT, Embedded &
Firmware Systems,
Solution Engineer

My Portfolio Website



What i’am doing in this project ?

- Worked on date: 1st July 2022 – 20th Agust 2022
- Made Responsive design portfolio web-based by using HTML, CSS, Javascript
- Resume also aviable in this link : https://cv-ats-mj.vercel.app/
- Accessible via link :https://mjmokhtar.github.io/portfoliomj/

Thanks!

More information

please contact me.

mjmmiat01@gmail.com

(+62)853 4941 6700

Muhammad Jumi'at Mokhtar

@mjmokhtar

mjmokhtar.cloud

Wani manimbai
Wani Manajuni

Berani melontar
Berani Terjun

MUHAMMAD
JUMI'AT MOKHTAR